

Loan Approval Prediction — Dataset Collection & Exploration

Project Phase 2 | Day 16 Task

1. Dataset Source

Dataset: Dream Housing Finance – Loan Prediction Dataset (a well-known publicly available dataset originating from the Analytics Vidhya 'Loan Prediction' practice problem, also widely mirrored on Kaggle as the 'Loan Approval Prediction Dataset').

2. Dataset Shape

Rows: 614 | Columns: 13

3. Column Overview & Data Types

Column	Data Type	Missing Values	Description
Loan_ID	str	0	Unique loan application ID
Gender	str	13	Applicant gender (Male/Female)
Married	str	3	Marital status (Y/N)
Dependents	str	15	Number of dependents (0,1,2,3+)
Education	str	0	Graduate / Not Graduate
Self_Employed	str	32	Self-employed (Y/N)
ApplicantIncome	int64	0	Applicant's income
CoapplicantIncome	float64	0	Co-applicant's income
LoanAmount	float64	22	Loan amount (in thousands)
Loan_Amount_Term	float64	14	Loan term (in months)
Credit_History	float64	50	Credit history meets guidelines (1/0)
Property_Area	str	0	Urban / Semiurban / Rural
Loan_Status	str	0	Target: Loan approved (Y/N)

4. Dataset Preview (First 5 Rows)

Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
LP001002	Male	No	0	Graduate	No	5849	0.0		360.0	1.0	Urban	Y
LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

5. Approval Distribution (Loan_Status)

Loan Status	Count	Percentage
Approved (Y)	422	68.7%
Rejected (N)	192	31.3%

The dataset is moderately imbalanced: about 68.7% of applications were approved, while 31.3% were rejected. This should be considered when training the machine learning model (e.g. using stratified sampling or class-weighting).

6. Missing Value Summary

Columns with missing values: Credit_History (50), Self_Employed (32), LoanAmount (22), Dependents (15), Loan_Amount_Term (14), Gender (13), Married (3).